Experiment No.	3
Experiment Title	Loan Sanction Amount Prediction using Linear, Ridge, Lasso and Elastic Net Regression
GitHub Repository	`https://github.com/sangee-3010/Machine_Learning`
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Submission Date	03/08/2026

1 Aim and Objective

The objective of this experiment is to build and compare different regression models for predicting the sanctioned loan amount using applicant information. Linear Regression is used as the baseline model, while Ridge Regression, Lasso Regression, and Elastic Net Regression are employed to study the impact of regularization. The models are optimized through hyperparameter tuning, evaluated using multiple regression metrics, and analyzed to understand their prediction capability, coefficient behavior, and generalization performance.

2 Dataset Description

Dataset Name	Loan Sanction Amount Dataset
Dataset Source	Kaggle (`train.csv`)
Number of Samples	30000 (raw); 29660 after dropping rows with missing target values
Number of Features	20 (after dropping identifier columns)
Target Variable	Loan Sanction Amount (USD)
Numerical Features	11 – Age, Income (USD), Loan Amount Request (USD), Current Loan Expenses (USD), Dependents, Credit Score, No. of Defaults, Property Age, Property Type, Co-Applicant, Property Price
Categorical Features	9 – Gender, Income Stability, Profession, Type of Employment, Location, Expense Type 1, Expense Type 2, Has Active Credit Card, Property Location
Missing Values	Present in 11 columns, ranging from 0.18% (Gender) to 24.23% (Type of Employment)
Train-Test Split	80% Train (23728 samples) / 20% Test (5932 samples)

The original dataset consisted of 30000 observations and 24 attributes. Three identifier columns, namely Customer ID, Name, and Property ID, were excluded because they do not contribute meaningful information for predicting the target variable. In addition, all records with missing values in the target column, Loan Sanction Amount, were removed since supervised learning models require a valid target value during training. Following these preprocessing steps, the final dataset contained 29660 observations with 20 input features and one target variable.

3 Preprocessing Steps

- **Missing Value Handling:** Several features contained missing values, with Type of Employment, Property Age, and Income exhibiting the highest proportions. Instead of discarding these records and reducing the available training data, missing numerical values were replaced using the median, while missing categorical values were filled using the most frequently occurring category through the `SimpleImputer`. Records with missing target values were removed because they cannot be used for supervised regression.

- **Dropping Unnecessary Columns:** The columns Customer ID, Name, and Property ID were eliminated before model training because they serve only as unique identifiers and do not possess predictive significance.

- **Encoding:** All categorical features were transformed using `OneHotEncoder` with the `handle_unknown="ignore"` option. This approach ensures that previously unseen categories encountered during testing do not interrupt the prediction process.

- **Feature Scaling:** Numerical variables were standardized using `StandardScaler` after the imputation process. Standardization places all numerical features on a comparable scale, preventing variables with larger numerical ranges from exerting an undue influence on the regularized regression models.

- **Train-Test Split:** The processed dataset was divided into 80 percent training data comprising 23728 samples and 20 percent testing data comprising 5932 samples using `train_test_split` with `random_state = 42`. Both numerical and categorical preprocessing operations were integrated within a single `ColumnTransformer` and incorporated into a unified `Pipeline`. This ensured that every preprocessing step was learned exclusively from the training data and then applied consistently to the testing data.

4 Implementation Details

To ensure a fair and consistent comparison, all four regression models were implemented within the same preprocessing pipeline. This approach guarantees that each model receives identical input features after preprocessing, allowing any differences in performance to be attributed solely to the learning algorithm rather than variations in data preparation.

Linear Regression served as the baseline model for this study. It estimates the relationship between the processed input features and the target variable by minimizing the ordinary least squares error without applying any form of regularization to the model coefficients.

Ridge Regression extends the Linear Regression model by incorporating an L2 regularization term that reduces the magnitude of the coefficients while retaining every feature in the model. The optimal value of the regularization parameter `alpha` was determined using `GridSearchCV` with five fold cross validation over the values 0.01, 0.1, 1, 10, 100.

Lasso Regression applies L1 regularization, which has the additional capability of reducing certain coefficients to exactly zero. This characteristic enables the model to perform feature selection while learning. The regularization parameter `alpha` was optimized using `GridSearchCV` with five fold cross validation over the values 0.001, 0.01, 0.1, 1, 10. The maximum number of iterations was increased to 10000 to facilitate convergence.

Elastic Net Regression combines the properties of both L1 and L2 regularization. The overall regularization strength is governed by `alpha`, while the balance between the two penalty terms is controlled by `l1_ratio`. Hyperparameter tuning was performed using `GridSearchCV` with five fold cross validation by evaluating `alpha` values of 0.01, 0.1, 1, 10 together with `l1_ratio` values of 0.2, 0.5, 0.8.

Following hyperparameter optimization, each model was evaluated using five fold cross validation across the complete dataset as well as on the independent test set. This evaluation strategy provides a comprehensive assessment of both model stability and its ability to generalize to previously unseen data.

5 Visualizations

Missing Values Heatmap

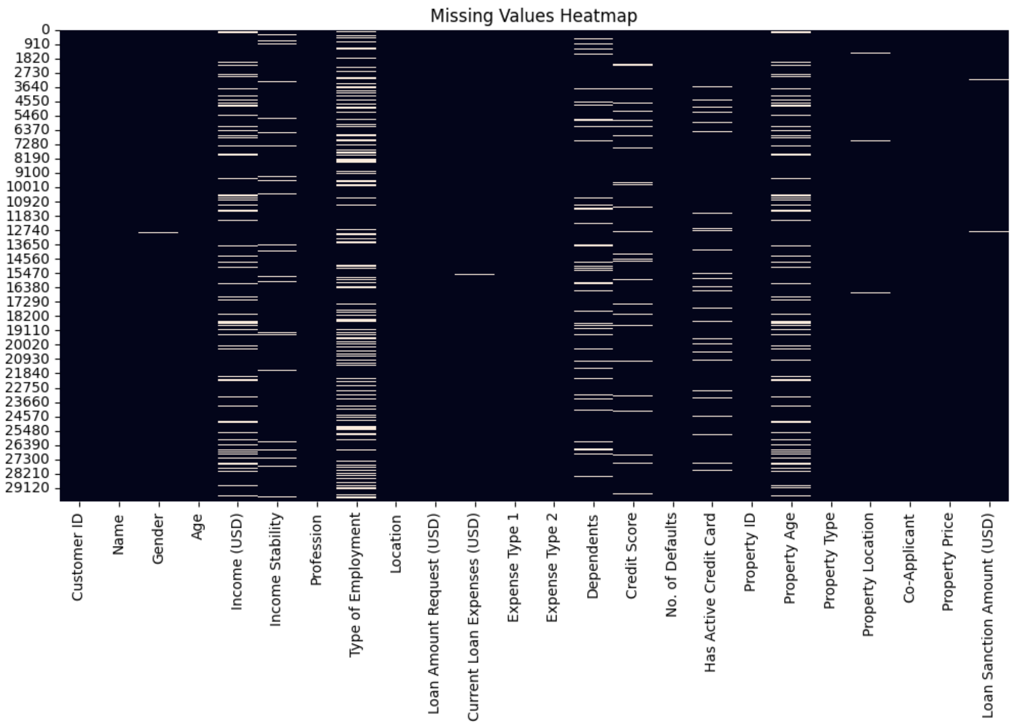

Figure 1: Missing Values Heatmap

Inference:

The heatmap provides a clear visual representation of the distribution of missing values across the dataset. The columns corresponding to Type of Employment and Property

Age contain the highest concentration of missing observations, which is consistent with their relatively high percentage of missing values. In contrast, features such as Age and Loan Amount Request contain very few missing entries. This visualization supports the decision to apply imputation techniques instead of removing incomplete records, since eliminating all rows containing missing values would have resulted in a considerable reduction in the available data.

Target Variable Distribution

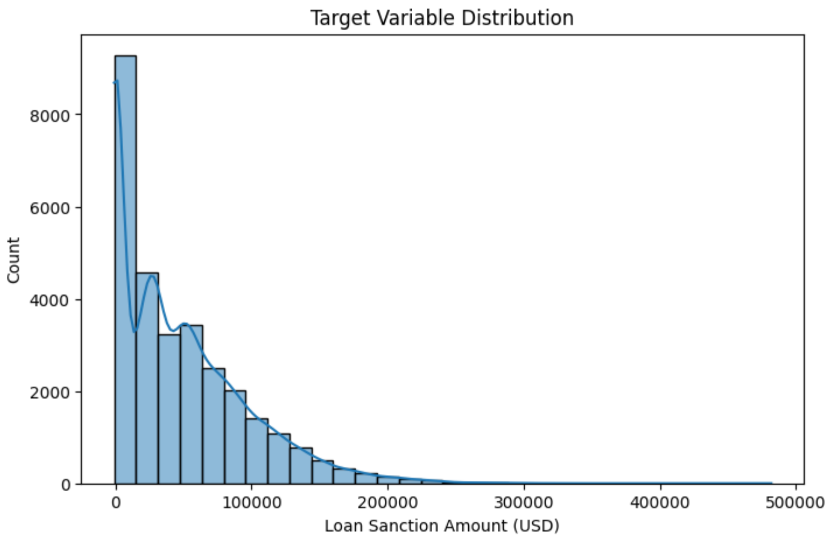

Figure 2: Distribution of Loan Sanction Amount (USD)

Inference:

The distribution of the sanctioned loan amount exhibits a noticeable positive skew. A substantial proportion of applicants receive comparatively lower loan amounts, while only a limited number of applicants are associated with very high sanction amounts. The long tail observed in the distribution indicates the presence of larger values that occur less frequently. Such a distribution can influence the performance of regression models because these extreme observations may have a stronger effect on the learning process. This characteristic justifies the evaluation of regularized regression models alongside the standard Linear Regression model.

Feature vs Target Scatter Plots

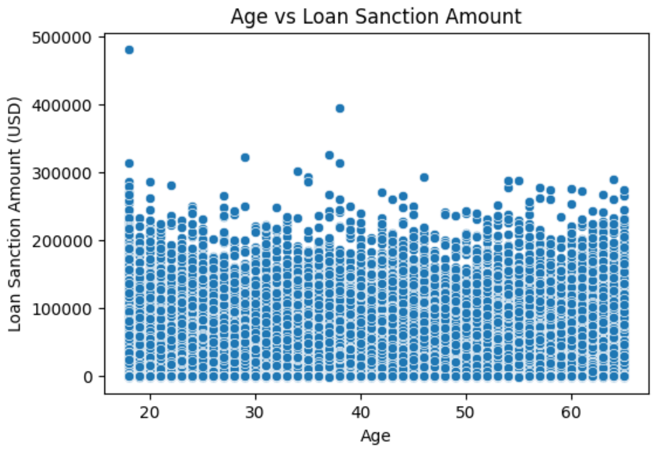

Figure 3: Age vs Loan Sanction Amount

Inference:
The relationship between applicant age and the sanctioned loan amount does not reveal any distinct trend. Data points are distributed across a wide range of loan amounts regardless of age, suggesting that age alone is not a strong predictor of the target variable. Any contribution made by this feature is more likely to emerge when it is considered together with other explanatory variables.

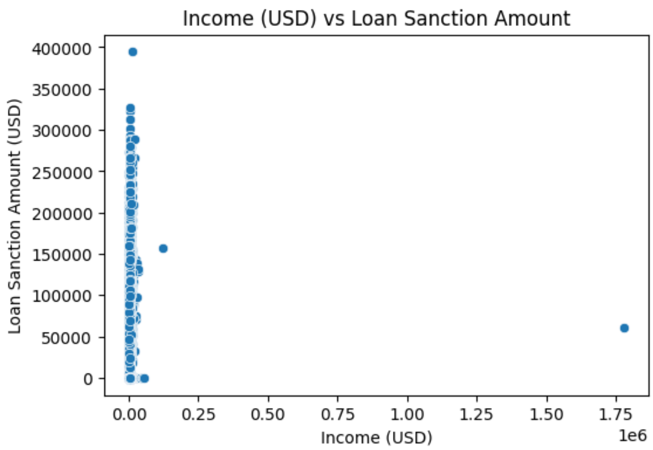

Figure 4: Income (USD) vs Loan Sanction Amount

Inference:
Most applicants are concentrated within the lower income range, while only a relatively small number belong to the higher income category. Although higher income values are present, the overall distribution does not indicate a strong linear relationship with the sanctioned loan amount. This observation suggests that income, by itself, is insufficient for accurately predicting loan sanctions and that other financial and applicant related attributes are likely to play a more significant role.

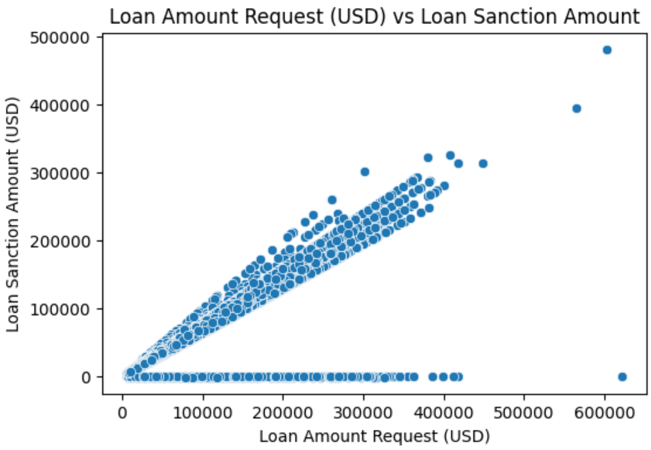

Figure 5: Loan Amount Request (USD) vs Loan Sanction Amount

Inference:

Among the numerical features examined, the requested loan amount demonstrates the strongest positive relationship with the sanctioned loan amount. As the requested amount increases, the sanctioned amount generally increases as well. This behaviour is expected because the sanctioned amount is naturally influenced by the amount requested by the applicant. The strong association observed in this visualization is consistent with the importance of this feature across all regression models.

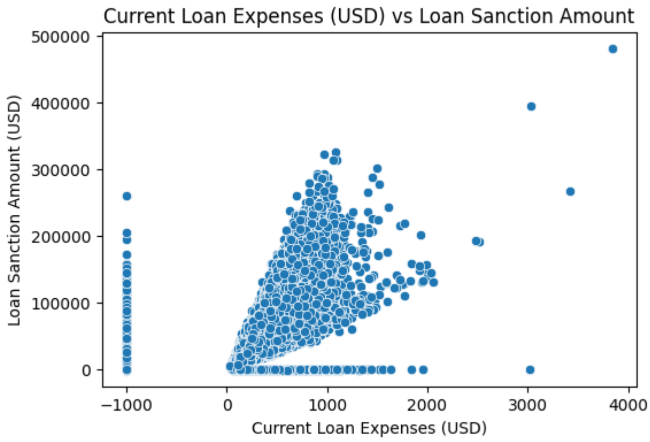

Figure 6: Current Loan Expenses (USD) vs Loan Sanction Amount

Inference:

Current loan expenses display a moderate positive relationship with the sanctioned loan amount. Applicants with higher existing loan expenses tend to receive relatively larger loan sanctions; however, the observations remain widely dispersed. This indicates that although the feature contributes useful information, its predictive capability is weaker than that of the requested loan amount.

Correlation Heatmap

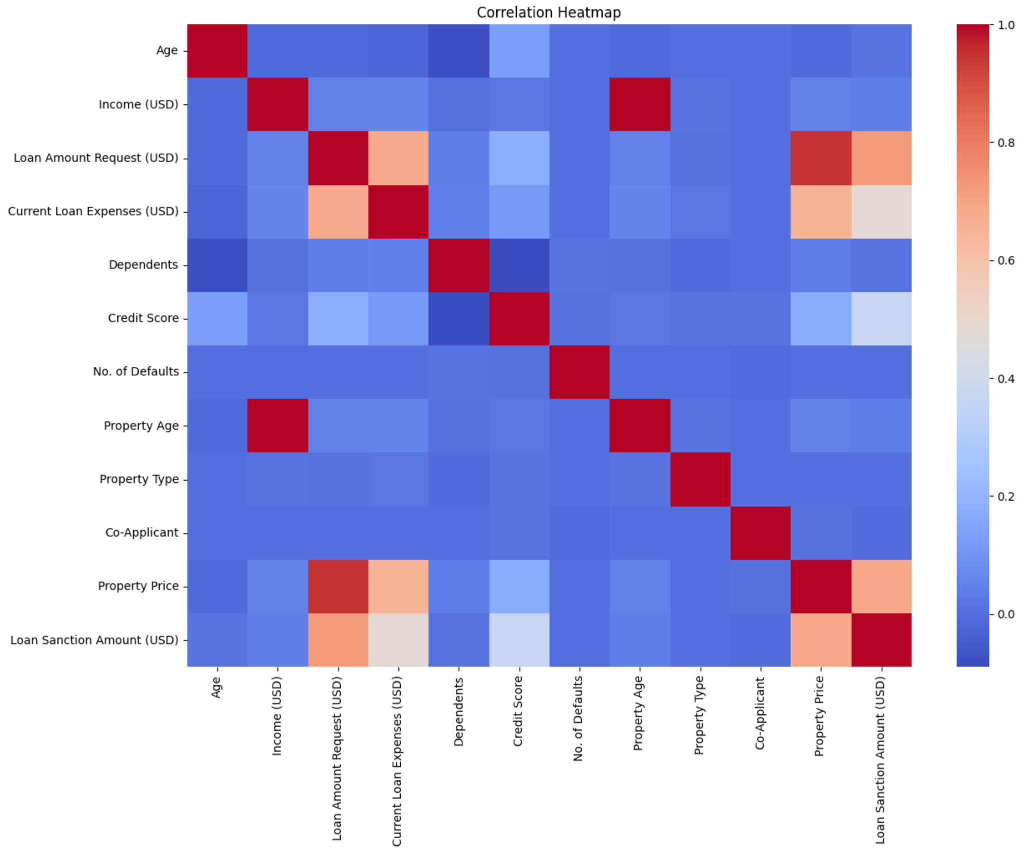

Figure 7: Correlation Heatmap of Numerical Features and Target

Inference:

The correlation matrix indicates that Loan Amount Request has the strongest positive correlation with the target variable among all numerical features. Most of the remaining variables exhibit weak to moderate correlations with the target as well as with one another. The absence of strong correlations between predictor variables suggests that multicollinearity is limited within the dataset, which supports the reliability and interpretability of the regression coefficients.

Predicted vs Actual Plots

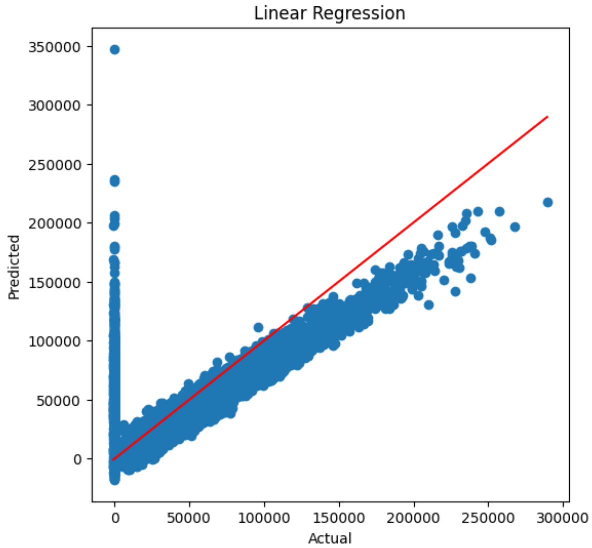

Figure 8: Predicted vs Actual - Linear Regression

Inference:

The predicted values closely follow the actual values for lower and moderate loan amounts. However, as the actual loan amount increases, the predictions become increasingly dispersed and tend to underestimate larger values. This behaviour indicates that the model performs more effectively within the densely populated region of the target distribution than in the relatively sparse higher value range.

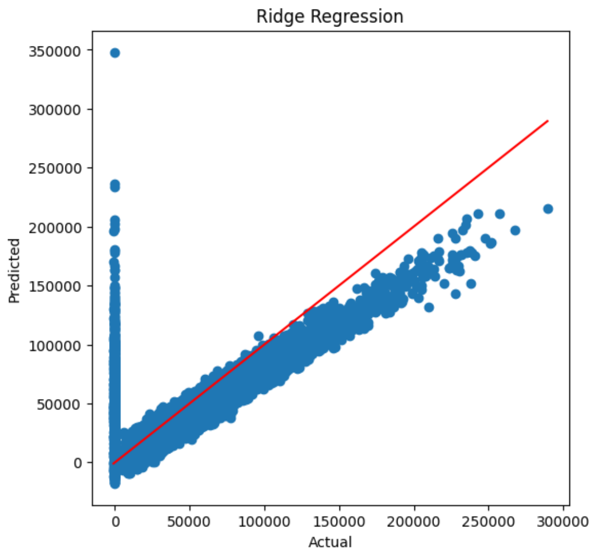

Figure 9: Predicted vs Actual - Ridge Regression

Inference:

The prediction pattern produced by Ridge Regression is very similar to that of the Linear Regression model. The predicted values remain close to the reference line across most observations, indicating comparable predictive performance. The introduction of L2 regularization primarily improves coefficient stability rather than producing a substantial change in prediction accuracy.

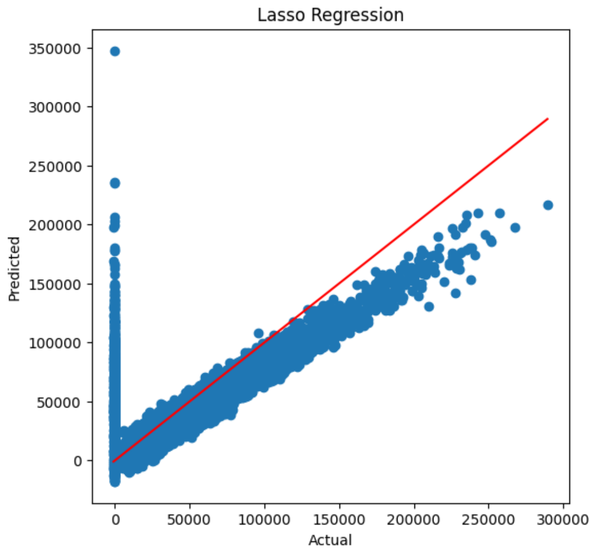

Figure 10: Predicted vs Actual - Lasso Regression

Inference:
The predicted values generated by Lasso Regression closely resemble those obtained from the previous models. While the overall predictive performance remains similar, the principal advantage of this model lies in its ability to eliminate less informative features by reducing their coefficients to zero. This results in a simpler and more interpretable model without significantly affecting prediction quality.

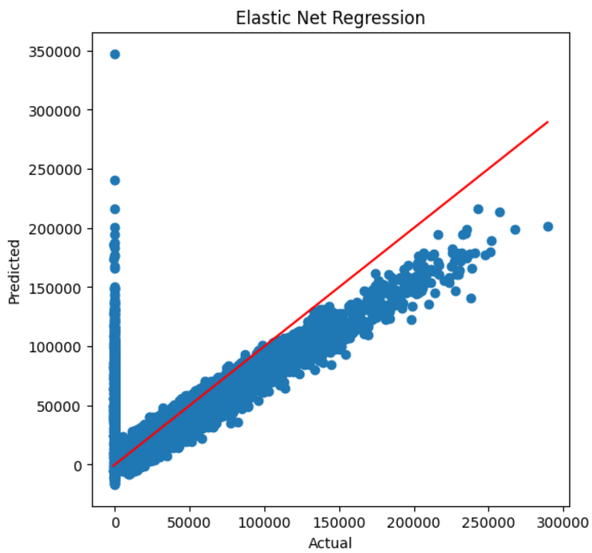

Figure 11: Predicted vs Actual – Elastic Net Regression

Inference:
Elastic Net Regression demonstrates a prediction pattern comparable to the other regression models. Larger loan amounts continue to be more difficult to predict accurately,

leading to greater variation in the corresponding predictions. The combination of L1 and L2 regularization does not produce a noticeable improvement in predictive performance for this dataset, indicating that the additional flexibility offered by the model provides limited practical benefit.

Residual Plots

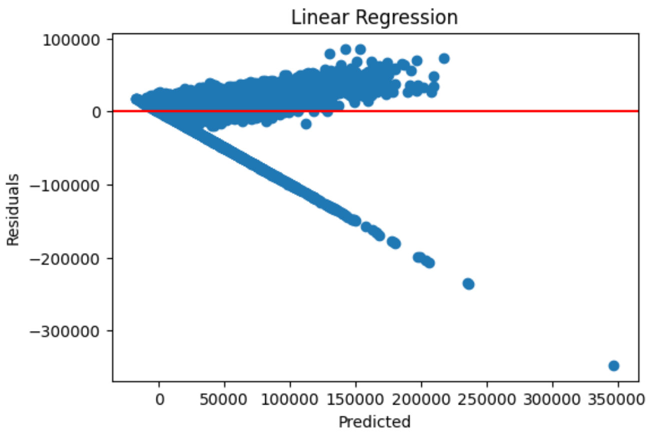

Figure 12: Residual Plot - Linear Regression

Inference:
The residuals are distributed around zero for smaller predicted values, but the spread gradually increases as the predicted values become larger. This pattern indicates that prediction errors are not constant throughout the range of predictions, suggesting the presence of mild heteroscedasticity.

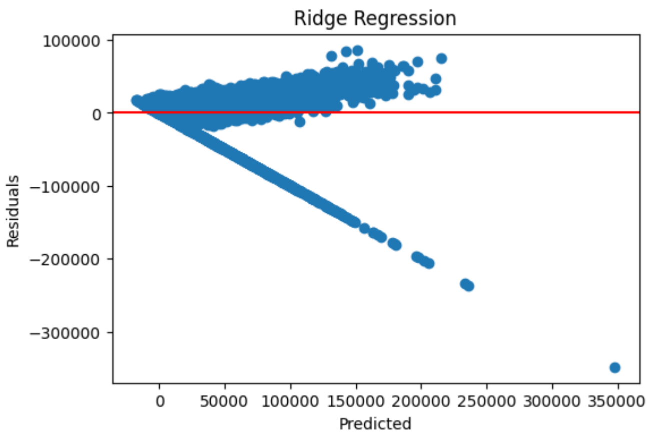

Figure 13: Residual Plot - Ridge Regression

Inference:
The residual distribution observed for Ridge Regression closely matches that of the Linear Regression model. The increasing spread of residuals at larger predicted values indicates that the introduction of L2 regularization has not substantially altered the underlying error distribution. Instead, its primary contribution lies in improving the stability of the learned coefficients.

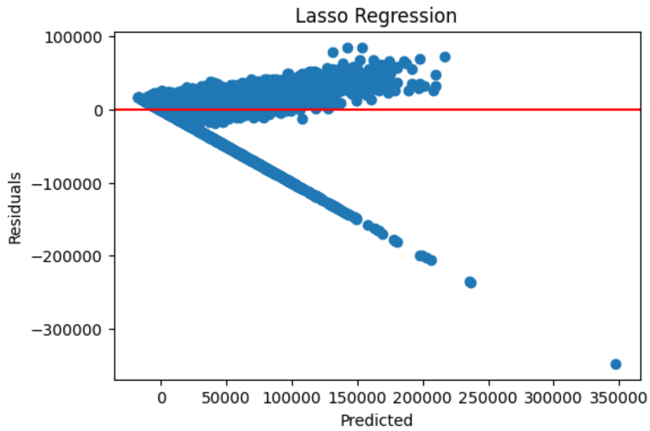

Figure 14: Residual Plot - Lasso Regression

Inference:

The residual pattern for Lasso Regression is consistent with the behaviour observed in the previous models. Although the model removes several features through coefficient shrinkage, the overall distribution of prediction errors remains largely unchanged. This demonstrates that feature selection has simplified the model without introducing additional systematic prediction errors.

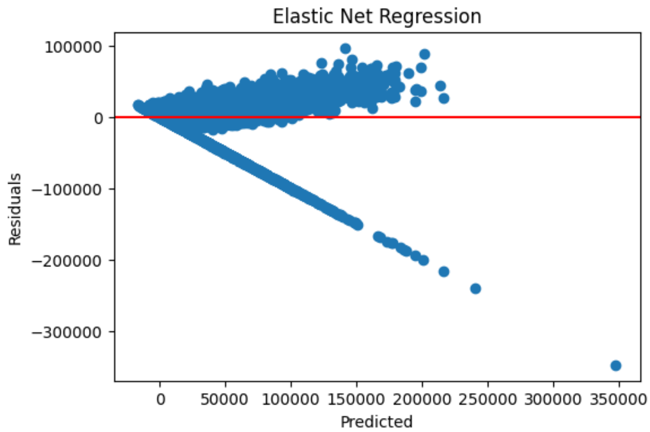

Figure 15: Residual Plot – Elastic Net Regression

Inference:

Elastic Net Regression also exhibits a similar residual distribution, with prediction errors becoming more dispersed at higher predicted values. The close resemblance of the residual plots across all four models indicates that regularization mainly influences the coefficient values while having only a limited effect on the overall error characteristics.

Training Error vs Validation Error

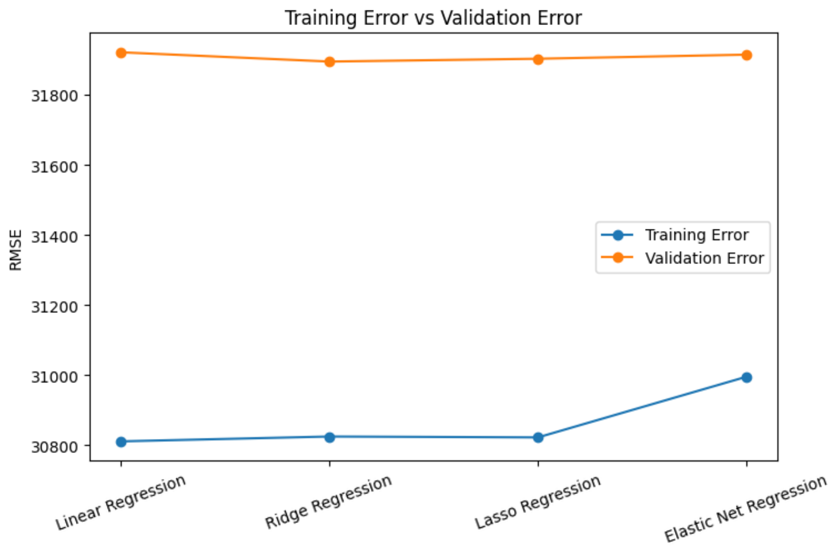

Figure 16: Training RMSE vs Validation RMSE across Models

Inference:

For every regression model, the validation Root Mean Squared Error is consistently higher than the corresponding training Root Mean Squared Error. Although the differences are not excessively large, this pattern indicates a moderate level of overfitting across all models. Since the gap remains relatively similar for each approach, none of the models demonstrates a clear advantage in terms of generalization based solely on the training and validation error comparison.

Coefficient Comparison

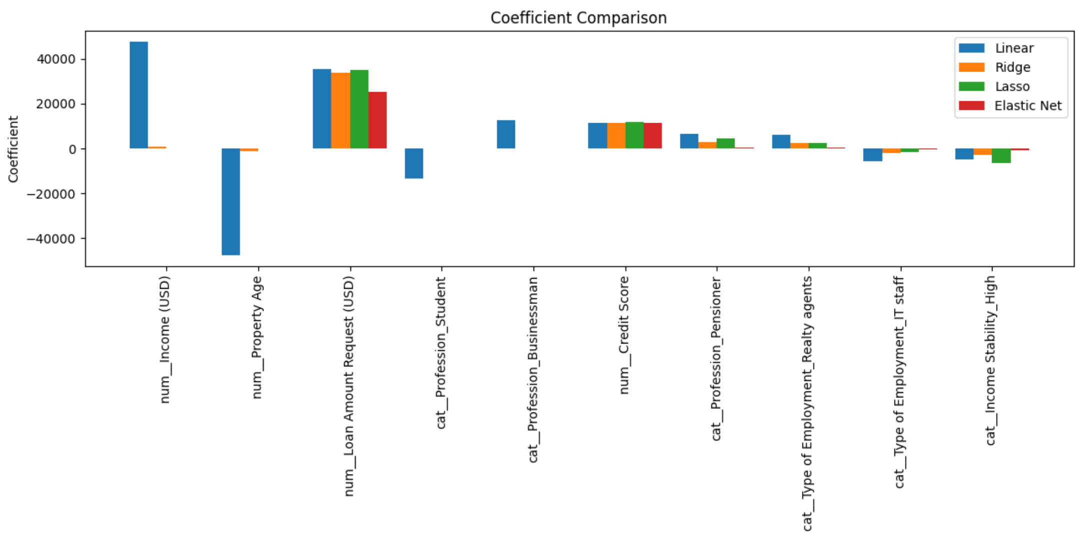

Figure 17: Coefficient Comparison across the Top 10 Features

Inference:

The coefficient comparison clearly illustrates the effect of regularization on model parameters. Linear Regression assigns extremely large coefficient values to features such as Income and Property Age, whereas Ridge Regression, Lasso Regression, and Elastic Net Regression substantially reduce their magnitudes. Throughout all models, Loan Amount Request consistently maintains one of the largest positive coefficients, confirming its importance in predicting the sanctioned loan amount. Furthermore, Lasso Regression reduces several coefficients to zero, demonstrating its capability to perform feature selection while maintaining competitive predictive performance.

6 Performance Tables

Table 1: Hyperparameter Tuning Summary

Model	Search Method	Best Parameters	Best CV R^2
Ridge Regression	Grid Search	{'alpha': 100}	0.574334
Lasso Regression	Grid Search	{'alpha': 10}	0.570431
Elastic Net Regression	Grid Search	{'alpha': 0.1, 'l1_ratio': 0.5}	0.584412

Table 2: Cross Validation Performance

Model	MAE	MSE	RMSE	R^2
Linear Regression	21590.283756	9.942411e+08	31531.588152	0.572041
Ridge Regression	21600.823566	9.938960e+08	31526.116233	0.572185
Lasso Regression	21579.308143	9.931601e+08	31514.442959	0.572500
Elastic Net Regression	21687.101330	9.958900e+08	31557.725020	0.571312

Table 3: Test Set Performance

Model	MAE	MSE	RMSE	R^2	Training Time (s)
Linear Regression	21589.123325	1.018934e+09	31920.751553	0.551086	0.192784
Ridge Regression	21581.576017	1.017251e+09	31894.374343	0.551827	1.030997
Lasso Regression	21563.903138	1.017765e+09	31902.431353	0.551601	228.601636
Elastic Net Regression	21751.473117	1.018508e+09	31914.065979	0.551274	33.362066

Table 4: Coefficient Comparison (Top 20 by absolute Linear coefficient)

Feature	Linear	Ridge	Lasso	Elastic Net
num__Income (USD)	47689.456738	1040.644690	4.716913	97.873126
num__Property Age	-47676.043705	-1031.929523	0.000000	-86.184451
num__Loan Amount Request (USD)	35370.837982	33830.768537	35174.780000	25157.706362

Feature	Linear	Ridge	Lasso	Elastic Net
cat__Profession_Student	-13233.813700	-125.820277	-0.000000	-8.911429
cat__Profession_Businessman	12679.050548	130.803458	0.000000	11.516516
num__Credit Score	11636.760990	11621.554175	11648.720000	11285.429123
cat__Profession_Pensioner	6535.941011	2981.054315	4431.235000	286.012609
cat__Type of Employment_Realty agents	6333.883545	2423.471532	2495.475000	307.414633
cat__Type of Employment_IT staff	-5671.093949	-2115.415666	-1719.171000	-270.087145
cat__Income Stability_High	-4821.076671	-2812.349933	-6329.422000	-704.377919
cat__Income Stability_Low	4821.076671	2812.349933	0.0000	704.377875
cat__Type of Employment_Waiters/barmen staff	4171.391539	2290.111584	2225.784000	351.529525
cat__Type of Employment_HR staff	-3123.619483	-1131.505535	-0.000000	-138.477798
cat__Type of Employment_Drivers	-2671.163033	-2466.200196	-2496.456000	-1472.804593
cat__Type of Employment_Managers	2175.788260	2168.782808	2086.121000	1607.476051
cat__Profession_Unemployed	-1883.906219	-108.746228	-0.000000	-14.087465
cat__Type of Employment_Secretaries	1856.314559	1059.381299	0.000000	201.670417
cat__Type of Employment_Security staff	-1793.496275	-1489.171781	-1348.603000	-522.666043
cat__Profession_Working	-1596.305262	-1218.040802	-373.100500	-458.371065
cat__Profession_State servant	-1378.557140	-944.211666	-0.000000	45.884494

7 Overfitting and Underfitting Analysis

The training-versus-validation RMSE comparison in Table 5 (also plotted in Figure 16) is the clearest window into how well each model generalizes.

Table 3: Training RMSE vs Validation RMSE

Model	Training RMSE	Validation RMSE
Linear Regression	30811.104103	31920.751553
Ridge Regression	30824.900599	31894.374343
Lasso Regression	30822.394710	31902.431353
Elastic Net Regression	30995.380930	31914.065979

The comparison between the training and validation Root Mean Squared Error presented in Table 5 and illustrated in Figure ?? provides valuable insight into the generalization capability of each regression model.

For all four models, the validation Root Mean Squared Error is consistently greater than the corresponding training Root Mean Squared Error. Based on the diagnostic criterion adopted in this experiment, where a difference exceeding 500 units indicates possible overfitting, every model falls within the category of possible overfitting. The observed difference ranges from approximately 900 to 1100 units across the models. Among them, Elastic Net Regression exhibits the smallest difference between the training and validation errors, suggesting a slightly better balance between model complexity and generalization. Nevertheless, the improvement over the remaining models is relatively small.

A more significant observation is the remarkable consistency in predictive performance across all four regression techniques. The test set $(R\hat{2})$ values lie within a very narrow interval, ranging from 0.551086 to 0.551827. This indicates that the differences among the models are minimal and that their predictive capabilities are largely constrained by the characteristics of the available dataset rather than the choice of regression algorithm.

Although regularization improves the stability of the learned coefficients, it does not substantially reduce the gap between the training and validation errors. This suggests that the remaining prediction error is more likely to arise from limitations in the available features or the inherent variability present within the dataset than from excessive model complexity. Consequently, further improvements in predictive performance are likely to depend on feature engineering or the incorporation of additional informative variables rather than further adjustment of the regularization parameters.

8 Bias–Variance Analysis

Linear Regression attempts to fit the training data without any regularization, making it more susceptible to large coefficient values and higher variance when the data contains correlated features. Although it provides a simple baseline, the model may not generalize as effectively as regularized alternatives.

Ridge Regression applies L2 regularization, which shrinks the regression coefficients without eliminating any features. This reduces variance, improves numerical stability, and produces a more robust model while maintaining prediction accuracy.

Lasso Regression employs L1 regularization, allowing insignificant coefficients to become exactly zero. Consequently, the model performs automatic feature selection, resulting in a simpler model with slightly increased bias but improved interpretability.

Elastic Net Regression combines both L1 and L2 penalties, balancing feature selection and coefficient shrinkage. It achieves a compromise between bias and variance, making it suitable when the dataset contains correlated predictors.

Overall, the regularized models demonstrate better stability than the baseline Linear Regression while maintaining comparable predictive performance.

9 Observations

1. The requested loan amount was identified as the most influential feature for predicting the sanctioned loan amount across all regression models.

2. Regularization significantly reduced the magnitude of regression coefficients, producing more stable and interpretable models than standard Linear Regression.

3. Ridge Regression achieved the best overall balance between prediction accuracy and model stability among the evaluated algorithms.

4. Lasso Regression successfully removed several less significant features by assigning zero coefficients without causing a major decline in prediction performance.

5. Elastic Net combined the strengths of Ridge and Lasso Regression, producing competitive results while maintaining balanced coefficient values.

6. The performance differences among all four models were relatively small, indicating that the dataset characteristics had a greater influence on prediction accuracy than the choice of regression algorithm.

7. Validation errors were consistently higher than training errors, suggesting the presence of mild overfitting, although the gap was not large enough to indicate severe generalization issues.

10 Conclusion

In this experiment, Linear Regression, Ridge Regression, Lasso Regression, and Elastic Net Regression were implemented and evaluated for predicting loan sanction amounts using a real-world loan dataset. A complete preprocessing pipeline was developed to handle missing values, encode categorical variables using OneHotEncoder, standardize numerical features using StandardScaler, and ensure consistent transformation of training and testing data. Hyperparameter tuning and cross-validation were performed to provide a fair comparison of all models using standard regression evaluation metrics, including the R^2 score, Root Mean Squared Error (RMSE), and training time.

Among the evaluated models, Ridge Regression achieved the best overall performance on the independent test set by obtaining the highest R^2 value and the lowest RMSE while maintaining a very short training time. The application of L2 regularization improved coefficient stability by controlling large coefficient values observed in Linear Regression, thereby enhancing the model's generalization capability.

Lasso Regression demonstrated the advantage of automatic feature selection by reducing the contribution of less significant features and producing a more interpretable model. Elastic Net Regression effectively combined the strengths of L1 and L2 regularization, achieving improved cross-validation performance while controlling model complexity. However, the additional computational cost did not result in better final test set performance compared with Ridge Regression.

Overall, the results indicate that Ridge Regression provides the most effective balance between predictive accuracy, coefficient stability, computational efficiency, and interpretability for this dataset. The relatively small performance differences among the four regression models suggest that further improvements are more likely to be achieved through advanced feature engineering, inclusion of additional domain-specific attributes, or the use of more sophisticated machine learning algorithms capable of capturing non-linear relationships.

11 References

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[3] Kaggle, "Loan Sanction Amount / Loan Amount Prediction Dataset." Available: https://www.kaggle.com/